

ANWAR AYOON

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PROFESSIONAL SUMMARY

Data analyst and applied machine learning practitioner with 3+ years of experience building predictive models, forecasting systems, and analytics dashboards across solar energy, government, and operations data. Comfortable across the full workflow SQL and Python pipelines, model development, and BI reporting in Tableau and Power BI.

SKILLS

Programming & Data: Python (Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow), SQL, R (dplyr, tidyr, Shiny)

Visualization & BI: Tableau, Power BI, Streamlit, Excel (Power Query, PivotTables, XLOOKUP), Plotly, Seaborn, Matplotlib

Statistics & ML: Regression, Classification, Random Forest, XGBoost, LightGBM, LASSO/Ridge, Time-Series Forecasting, A/B Testing, Hypothesis Testing, SHAP

Deep Learning: CNNs, Semantic Segmentation, Model Quantization & Pruning, Edge Deployment

Tools & Platforms: Git, Jupyter, AWS (S3, EC2), SQLite/PostgreSQL, Microsoft Project

EXPERIENCE

IT Technician | Data & Operations Analytics

Mar 2025 – Present

[TechCyte — Iowa State University Book Store](#) | Ames, Iowa

- Built Excel-based operational dashboards tracking daily repair volume, turnaround time, and customer satisfaction, giving managers a single view of service performance across check-in, repair, and check-out stages.
- Analyzed service funnel data to identify workflow bottlenecks, segmenting recurring issue types to prioritize process improvements and reduce resolution delays.
- Translated raw service data into concise summaries and recommendations for non-technical managers, supporting faster staffing and workflow decisions.
- Maintain documentation for recurring technical issues and support system updates across the service workflow.

Data Repository Intern

Dec 2022 – May 2023

[Ministry of Cooperation](#) | Indore, India

- Designed SQL workflows to digitize and structure cooperative financial records, improving audit accuracy and cutting query and retrieval time on legacy data.
- Built SQL-based migration pipelines to preserve legacy records at risk of loss, converting paper and semi-structured sources into queryable relational databases.
- Standardized bilingual data entry and validation rules across archival records, creating searchable digital records and improving data reliability for downstream reporting.
- Defined long-term data governance rules covering schema consistency, validation, and access, improving data quality and stakeholder usability across teams.

Junior Data Scientist

Mar 2022 – Mar 2023

[PV Diagnostics — promoted from Data Science Intern \(Jan 2022\)](#) | Indore, India

- Built Therma Magic, a thermal-imaging analysis tool that detects overheating faults in PV panels, MCBs, and transformers by computing per-component thermal flare and overlaying it on YOLO-based bounding boxes of the underlying equipment.
- Developed an electroluminescence (EL) correlation workflow using IR camera imagery to detect microcracks in PV cells, giving field engineers a repeatable way to catch defects invisible in standard inspections.
- Built an IV-Load (IVL) forecasting pipeline that correlated plant IVL data with NASA solar-radiation datasets, then trained an LSTM on combined weather and radiation inputs to forecast future radiation and predict plant yield.
- Built predictive models and automated dashboards tracking plant performance, degradation patterns, and operational KPIs, replacing ad hoc reporting for the engineering team and feeding predictions into preventive-maintenance workflows.

PROJECTS

Personal Inflation Analytics Dashboard | Live Demo & GitHub

June 2026

- Building an interactive analytics dashboard using BLS Consumer Price Index data that lets users calculate personalized inflation rates by weighting their own spending categories against national trends.
- Designed an end-to-end Python pipeline (BLS API ingestion → SQLite → Streamlit deployment) processing 10+ years of CPI category data across 200+ item groups.

- Implemented interactive Plotly visualizations with category filtering, year-over-year comparison, and side-by-side personal vs. national inflation views, deployed on Streamlit Community Cloud.

Financial Well-Being Analytics | CFPB Survey

Aug 2025 – Sept 2025

- Built Python data pipelines to clean, transform, and analyze 6,300+ CFPB survey responses, enabling statistical modeling and feature engineering for financial well-being prediction.
- Improved model explanatory power from $R^2 = 0.14$ to $R^2 = 0.59$ by progressively incorporating economic and behavioral factors into survey-weighted regression models.
- Applied LASSO regression to reduce a high-dimensional feature set to ~60 key predictors, balancing interpretability and performance.
- Trained a tuned XGBoost regression model with interaction features, achieving 0.64 test R^2 and validating feature importance using SHAP values.

Fast-SCNN for Real-Time Road Segmentation | Autonomous Vehicle Edge Deployment

May 2025 – Jul 2025

- Designed and trained a Fast-SCNN semantic segmentation model in PyTorch for autonomous driving, building an end-to-end pipeline including data preprocessing, model training, evaluation, and edge deployment.
- Improved training stability and reduced overfitting through data augmentation, batch normalization, and learning-rate scheduling across training epochs.
- Applied quantization and structured pruning to compress the model, achieving a 3× reduction in inference latency while maintaining comparable segmentation accuracy.
- Deployed the optimized model on an NVIDIA Jetson Nano for real-time inference under embedded compute constraints suitable for autonomous vehicle applications.

LinkedIn Job Market Analysis | [Medium Blog](#)

April 2025

- Analyzed 124K+ LinkedIn job postings using XGBoost, Random Forest, and LightGBM to identify key salary drivers, finding experience level as the most predictive factor across industries.
- Applied SHAP-based model interpretability and industry-level segmentation to quantify the variable impact of experience across tech, healthcare, and finance.
- Visualized remote vs. on-site salary disparities using Seaborn and Plotly, finding remote jobs offered up to 30% higher average pay in knowledge industries across all experience levels.
- Conducted a deep-dive into benefit transparency (401(k), childcare, student loan support), uncovering strong correlations between explicit salary information and application rates (+50% engagement in some cases).

OCR + Object Detection Tool | Open-Source

Jun 2023 – Aug 2023

- Built an OCR and object detection tool in Python using PyTesseract and YOLOv5, achieving 95% accuracy on text and symbol extraction for medical equipment.
- Designed a custom labeling system to support operational efficiency and security requirements in a specialized medical-imaging context.

EDUCATION

Ivy College of Business, Iowa State University

Expected May 2027

Ames, Iowa | Master of Science in Information Systems | **4.0 GPA**

Medicaps University of Science & Technology

Sep 2019 – Jul 2023

Indore, India | Bachelor of Technology in Computer Science & Engineering | **3.5 GPA**

CERTIFICATIONS

Google Data Analytics Professional Certificate | Coursera

Mathematics for Machine Learning Specialization | Coursera

AWS Academy Cloud Foundations (S3, EC2) | AWS Academy

HONORS & PUBLICATIONS

TurfIt: Turf Reservation Systems — Published in Holkar Science College Journal, Apr 2023.

Smart India Hackathon (SIH) 2022 — Top-5 nationally for building a college ranking application for prospective students.